

SUMMARY

Computer Science graduate and Systems-driven Full-stack engineer, focused on choosing the right tool for each problem. Proven ability to build scalable web applications and APIs using SQL and NoSQL databases. Skilled at decomposing complex problems to deliver maintainable, high-quality code. Leverages AI for design validation, requirement refinement, and accelerated development. Seeking a collaborative team committed to deliberate architecture and long-term impact.

PROFESSIONAL PROJECTS

WhatsApp Product Verification (WPV) | [WPV Demo](#)

- ❖ Developed a Node.js/TypeScript anti-counterfeiting platform processing millions of product verifications via WhatsApp, implementing enterprise-grade security (JWT authentication, HMAC-SHA256 webhook validation, row-level database locking).
- ❖ Eliminated race-condition fraud using PostgreSQL locking (FOR UPDATE) and achieved sub-24ms p95 response times through indexing, query optimization, and connection pooling.
- ❖ Reduced API latency by 92% (from 2,000ms industry standard to 24ms) while load-testing to 200 concurrent users with 100% uptime and zero failed requests.

Rental Management System | [Admin Demo](#) | [Landlord Demo](#) | [Tenant Demo](#) | [Caretaker Demo](#)

- ❖ **Admin Dashboard** – Built admin reporting stack (PostgreSQL materialized views + Redis TTL caching) and BullMQ async PDF/Excel exports with status polling, secured by RS256 JWT with refresh revocation and audit logs.
- ❖ **Landlord Dashboard** – Built multi-tenant landlord APIs (Prisma ownership filters, parallel aggregates, composite indexes, SCAN-based cache invalidation) and rent-cycle accounting (monthly charges, allocations, reconciliations) replacing simple balance snapshots.
- ❖ **Tenant Portal** – Built race-safe M-Pesa reconciliation (PostgreSQL advisory locks + Serializable transactions + unique receipt constraints) and Lipa Mdogo Mdogo instalment schedules with opt-in conflict guards to prevent duplicate credits and double-billing.
- ❖ **Caretaker Portal** – Built property-scoped complaint APIs (idempotent assignment, status transitions, signed media) and Redis-backed 24hr SMS cooldowns with BullMQ to cut duplicate follow-ups.

Fitness Tracker | [Fitness Tracker Repository](#) | [Fitness Tracker Demo](#)

- ❖ Architected a fitness platform with decoupled microservices (React, Node.js, Python Flask) enabling workout, nutrition, and mental health tracking with ML-powered recommendations.
- ❖ Engineered multi-layer LRU caching with bounded 100-entry Flask cache and 4-minute Node.js TTL, achieving 70% cache hit rates and eliminating cold-start timeouts via async model loading on serverless infrastructure.

EDUCATION

Bachelor of Science in Computer Science | Chuka University | Graduated in Nov 2025

- ❖ **Relevant Coursework:** Software Engineering, User Interface Design, Distributed Systems, Decision Support Systems, Component Based Architecture, Software Project Management, Artificial Intelligence, Data Structures.

TECHNICAL STACK

- ❖ **Frontend & Languages:** React (Vite), Next.js (App Router), TypeScript, JavaScript, Vue, Tailwind CSS, TanStack Query, Material UI, ShadcnUi, Flutter, Python.
- ❖ **Backend & Databases:** Node.js, Express.js, Winston, Fast API, Laravel, PostgreSQL, MongoDB, Redis, BullMQ, Prisma, Zod.
- ❖ **DevOps & Tools:** GitHub Actions (CI/CD), Docker, Jest, Playwright, Vercel, Sentry, Git, GitHub, Render, Resend, Linux, CLI, Supertest, k6, Ngrok, Pino.
- ❖ **Core Competencies:** Context Engineering, Systems Thinking, AI-Augmented Development, Adaptability, Proactive Learning, Problem Solving, Teamwork, Time Management.

CERTIFICATION

- ❖ Docker Bootcamp for ML, GenAI and Agentic AI.
- ❖ Prompt Engineering with Amazon Q Developer.
- ❖ Cybersecurity & Emerging Technologies Awareness.